ANNEXES
1 Convolutional neurons' gradient descent

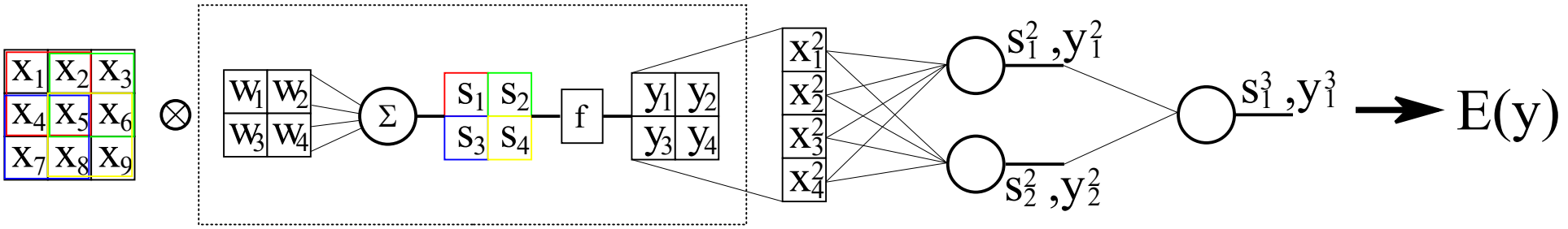
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- **Gradient descent of convolutional neurons**

  - Let's start with a very simple network
    - A single convolutional network with 4 weights (kernel of 2x2)
    - An image of 3x3 pixels, without padding

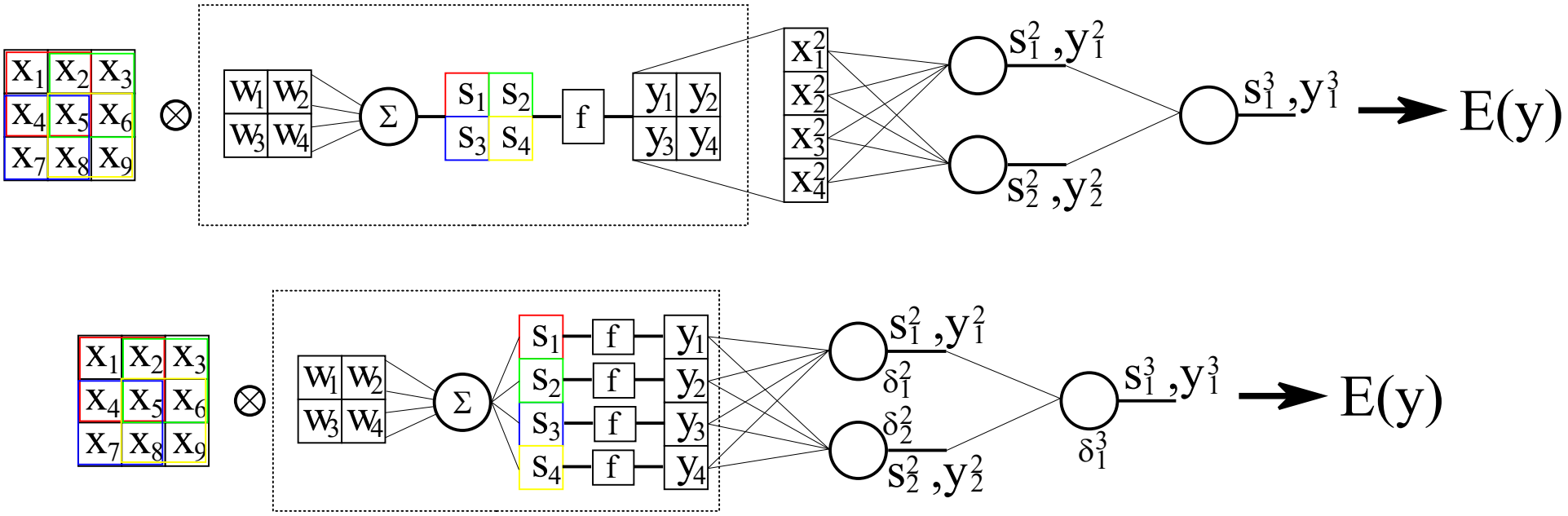
      → the convolution image has a size of 2x2 pixels

      → The first hidden layer has a 4-elements vector as input
    - A fully connected network with two layers



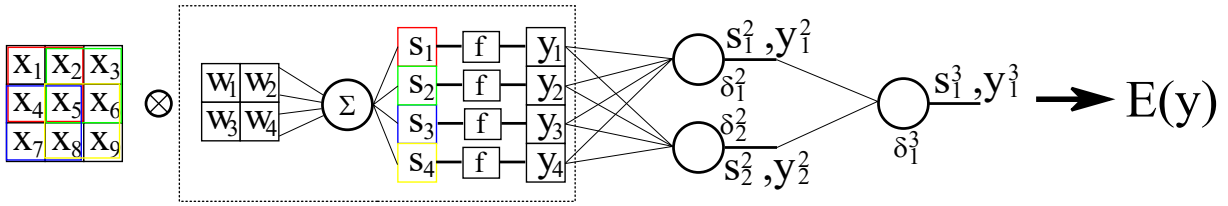
ANNEXES
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- Multiple "paths"



- We still want to calculate $\frac{\partial E(y)}{\partial w_i}$
- We know the δ of fully connected neurons

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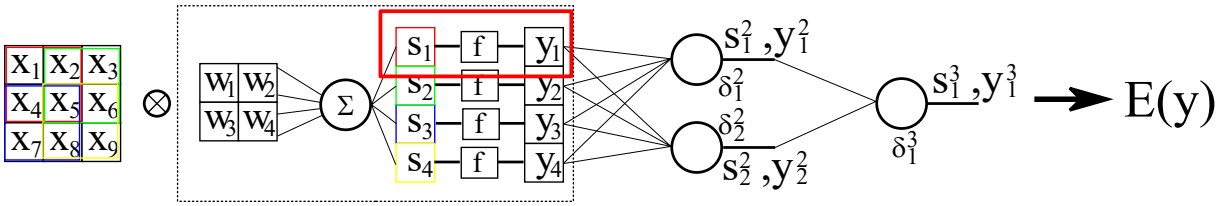
- Differentiation Rule for Composite Functions (via y1, y2, y3 and y4)

$$\frac{\partial E\left(y_1^3\right)}{\partial w_i} = \frac{\partial E\left(y_1^3\right)}{\partial y_1} \cdot \frac{\partial y_1}{\partial s_1} \cdot \frac{\partial s_1}{\partial w_i} + \frac{\partial E\left(y_1^3\right)}{\partial y_2} \cdot \frac{\partial y_2}{\partial s_2} \cdot \frac{\partial s_2}{\partial w_i}$$

$$+ \frac{\partial E\left(y_1^3\right)}{\partial y_3} \cdot \frac{\partial y_3}{\partial s_3} \cdot \frac{\partial s_3}{\partial w_i} + \frac{\partial E\left(y_1^3\right)}{\partial y_4} \cdot \frac{\partial y_4}{\partial s_4} \cdot \frac{\partial s_4}{\partial w_i}$$

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1 Convolutional neurons' gradient descent

x_1 w_1	x_2 w_2	0
x_4 w_3	x_5 w_4	0
0	0	0



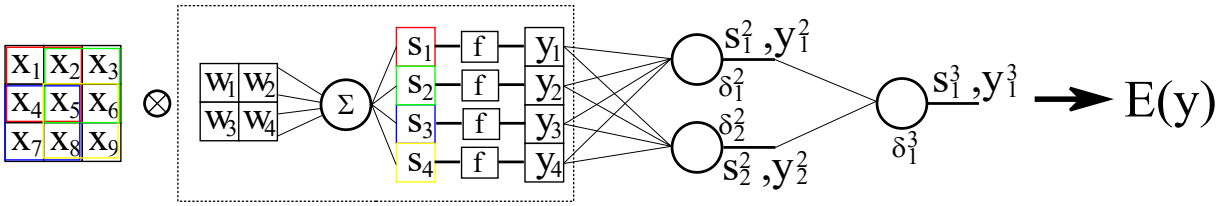
- Let's study one of them: $\frac{\partial E(y_1^3)}{\partial y_1} \cdot \frac{\partial y_1}{\partial s_1} \cdot \frac{\partial s_1}{\partial w_i}$

$$\sum_{k\in[1,n]} -\delta_k^2 \cdot w_{k1}^2 \cdot f'(s_1) \cdot \frac{\partial s_1}{\partial w_i}$$

$$\frac{\partial s_1}{\partial w_i} = \frac{\partial x_1.w_1 + x_2.w_2 + x_4.w_3 + x_5.w_4}{\partial w_i}$$

$$\frac{\partial s_1}{\partial w_1} = x_1 \quad \frac{\partial s_1}{\partial w_2} = x_2$$
$$\frac{\partial s_1}{\partial w_3} = x_4 \quad \frac{\partial s_1}{\partial w_4} = x_5$$

ANNEXES
1 Convolutional neurons' gradient descent



- Note: we must consider relations between weights w, position s and input x

x1 w1	x2 w2	0
x4 w3	x5 w4	0
0	0	0

$$\frac{\partial s_1}{\partial w_1} = x_1 \quad \frac{\partial s_1}{\partial w_2} = x_2 \quad \frac{\partial s_1}{\partial w_3} = x_4 \quad \frac{\partial s_1}{\partial w_4} = x_5$$

0	x2 w1	x3 w2
0	x5 w3	x6 w4
0	0	0

$$\frac{\partial s_2}{\partial w_1} = x_2 \quad \frac{\partial s_2}{\partial w_2} = x_3 \quad \frac{\partial s_2}{\partial w_3} = x_5 \quad \frac{\partial s_2}{\partial w_4} = x_6$$

0	0	0
x4 w1	x5 w2	0
x7 w3	x8 w4	0

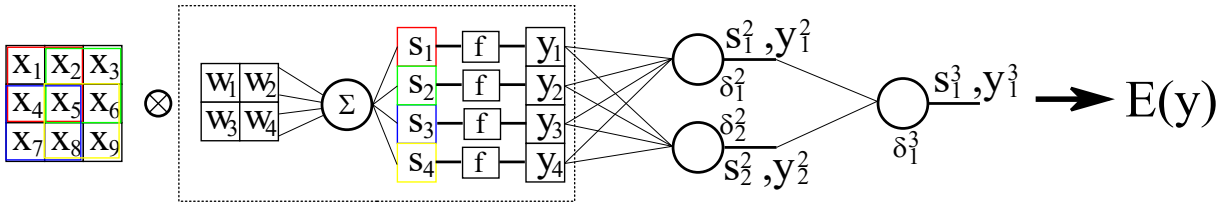
$$\frac{\partial s_3}{\partial w_1} = x_4 \quad \frac{\partial s_3}{\partial w_2} = x_5 \quad \frac{\partial s_3}{\partial w_3} = x_7 \quad \frac{\partial s_3}{\partial w_4} = x_8$$

0	0	0
0	x5 w1	x6 w2
0	x8 w3	x9 w4

$$\frac{\partial s_4}{\partial w_1} = x_5 \quad \frac{\partial s_4}{\partial w_2} = x_6 \quad \frac{\partial s_4}{\partial w_3} = x_8 \quad \frac{\partial s_4}{\partial w_4} = x_9$$

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x_1 w_1	x_2 w_1	x_3
x_4 w_1	x_5 w_1	x_6
x_7	x_8	x_9



- We replace the terms in the equation:

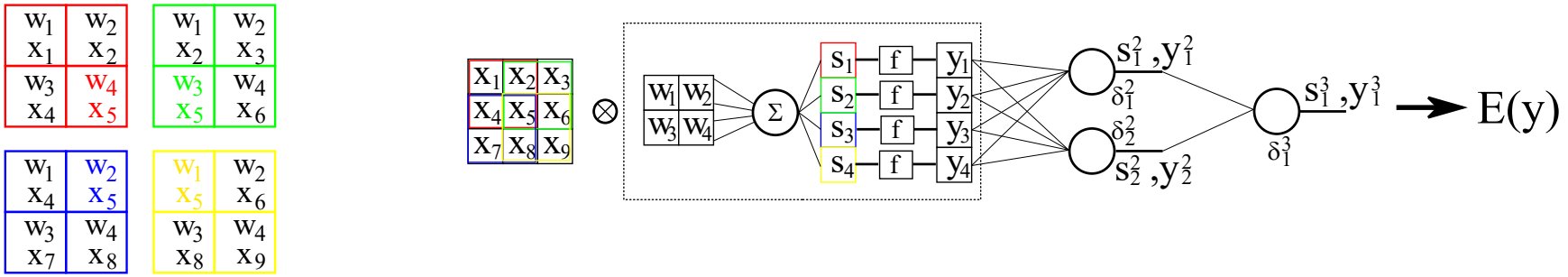
$$\frac{\partial E(y_1^3)}{\partial w_1} = -\underline{x_1 . f'(s_1) . \sum_{k\in[1,n]} \delta_k^2 \; . \; w_{k1}^2} - \underline{x_2 . f'(s_2) . \sum_{k\in[1,n]} \delta_k^2 \; . \; w_{k2}^2}$$

$$-\underline{x_4 . f'(s_3) . \sum_{k\in[1,n]} \delta_k^2 \; . \; w_{k3}^2} - \underline{x_5 . f'(s_4) . \sum_{k\in[1,n]} \delta_k^2 \; . \; w_{k4}^2}$$

$$\frac{\partial E(y_1^3)}{\partial w_1} = -(\underline{x_1 . \delta_1^1} + \underline{x_2 . \delta_2^1} + \underline{x_4 . \delta_3^1} + \underline{x_5 . \delta_4^1})$$

As many terms than pixels in the output image

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- Last step before backpropagating to previous layer:
  - Calculate $\frac{\partial E(y_1^3)}{\partial x_i}$

$$\frac{\partial E(y_1^3)}{\partial x_5} = \frac{\partial E(y_1^3)}{\partial s_1} \times \frac{\partial s_1}{\partial x_5} + \frac{\partial E(y_1^3)}{\partial s_2} \times \frac{\partial s_2}{\partial x_5} + \frac{\partial E(y_1^3)}{\partial s_3} \times \frac{\partial s_3}{\partial x_5} + \frac{\partial E(y_1^3)}{\partial s_4} \times \frac{\partial s_4}{\partial x_5}$$

δ_1^1

$$\frac{\partial x_1 . w_1 + x2 . w_2 + x_4 . w_3 + x_5 . w_4}{\partial x_5}$$

$$\frac{\partial E(y_1^3)}{\partial x_5} = -(w_4 . \delta_1^1 + w_3 . \delta_2^1 + w_2 . \delta_3^1 + w_1 . \delta_4^1)$$

As many terms than pixels in the kerne that Will pass over x_i